

# Stats 201A: Research Design, Sampling and Analysis

## Syllabus (Fall 2026)

**BruinLearn** homepage: <https://bruinlearn.ucla.edu/>

**Web Resources:** <http://www.stat.ucla.edu/~hqxu/stat201A/>

**Time/Place:** TR 11-12:15pm / Royce Hall 162 **Discussion:** W 10-10:50am/ Boelter Hall 5440

**Instructor:** Hongquan Xu <hqxu@stat.ucla.edu>; **TA:** Bradley Tang

**Office Hour:** T 2-2:50pm in MS 8955 or by appointment.

**Textbooks** (1-2)/References (3-6):

1. C. F. J. Wu and M. S. Hamada (2021). [Experiments: Planning, Analysis and Optimization](#), 3rd ed., Wiley. Library has E-book.
2. S. K. Thompson (2012). [Sampling](#), 3rd ed., Wiley. Library has E-book.
3. D. C. Montgomery (2005-2020). [Design and Analysis of Experiments](#), 6-10th ed., Wiley.
4. T. Lumley (2010). "[Complex Surveys: A Guide to Analysis Using R](#)", Wiley.
5. S. Sheather (2009) "[A Modern Approach to Regression with R](#)", Springer. Library has E-book.
6. J. J. Faraway (2005-2014). "[Linear Models with R](#)," Chapman & Hall.

**Overview:** We will spend 6-7 weeks on experimental design and 3-4 weeks on sampling. Tentative topics include

- **Research Design:** Basic principle, ANOVA, blocked designs, factorial designs, fractional factorial designs, response surface methods, computer experiments and space-filling designs. [Wu and Hamada (Ch. 1-5, 8, 10, 14) and notes]
- **Sampling:** Unequal Probability Sampling, Regression Estimation, Stratified Sampling, Cluster and Systematic Sampling, Multistage designs. [Thompson (Ch. 1-6, 8, 11-13)]

### Grading Policy:

- Attendance and polls/quizzes via iClicker (25%: including evaluation of peer presentations)
- Homework (25%) **No Late Homework.**
- Group project presentation (25%: peer evaluation) **Week 10 or Wednesday, December 9 (8am-11am).**
- Final group project report (25%) due **Saturday, December 12.**

**Group Project:** You are encouraged to work with other 1-2 students. The whole group hand in one written report. (a) Design an experiment, collect data and perform analysis, (b) Find 3 published experiments/data, redo the analysis (with improved results), (c) Other ideas (get approval first).

### iClicker Student Mobile App/Web App :

See "Students: Get Started with iClicker Cloud" at <https://coe.bruinlearn.ucla.edu/academic-tech-tools/iclicker-cloud/>.

### Statistical Software:

You are expected to be familiar with one statistical software (e.g., R, SAS, Stata, etc.). You are expected to know how to perform linear regression and make various plots. R is a language and environment for statistical computing and graphics. You can download R and get information from the R Home Page (<http://www.r-project.org/>).

**Academic Integrity:** As a student and member of the University community, you are here to get an education and are, therefore, expected to demonstrate integrity in your academic endeavors. All students must uphold

University of California Standards of Student Conduct as administered by the Office of the Dean of Students. Students are subject to disciplinary action for several types of misconduct, including but not limited to: cheating, multiple submissions, plagiarism, prohibited collaboration, facilitating academic dishonesty, or knowingly furnishing false information. You may have assignments or projects in which you work with a partner or with a group. For example, you are welcome, and even encouraged, to work with others to solve homework problems. Even though you are working together, the assignment you submit for a grade must be **IN YOUR OWN WORDS**, unless you receive specific instructions to the contrary. For more information about academic integrity, please go to <http://www.deanofstudents.ucla.edu>.